

ECE 532/COS 572/MAE 572/ROB 532
Safety-Critical Robotics and AI
 Problem Set 1

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Problem 1: Reachability Analysis

Conventions. For the autonomous system, $x_{x_0, t_0}(\cdot)$ denotes the unique trajectory that starts from x_0 at time t_0 . For the controlled system, $x_{x_0, t_0}^u(\cdot)$ denotes the trajectory generated by the control signal $u(\cdot) \in \mathbb{U}_{t_0}^t$, the set of admissible control signals on $[t_0, t]$, which we assume to be nonempty. Trajectories are continuous in time. Since “ $A \subseteq B$ ” means “ $x_0 \in A \Rightarrow x_0 \in B$ ”, each proof below takes an arbitrary point of the left-hand set and exhibits the witnesses required by the definition of the right-hand set, and each counterexample exhibits a point that lies in the left-hand set but not in the right-hand set.

Part 1: autonomous system $\dot{x} = f(x)$

- (a) **True.** Let $x_0 \in \overleftarrow{\mathcal{R}}_t(\mathcal{T})$, i.e. $x_{x_0, t_0}(t) \in \mathcal{T}$. Choose $\tau := t$. Then $\tau \in [t_0, t]$ and $x_{x_0, t_0}(\tau) \in \mathcal{T}$, which is exactly the defining condition of $\overleftarrow{\mathcal{R}}_{[t_0, t]}(\mathcal{T})$; hence $x_0 \in \overleftarrow{\mathcal{R}}_{[t_0, t]}(\mathcal{T})$. In words, the tube collects the states that visit $\mathcal{T}$ at *some* time in the horizon, in particular those that are in $\mathcal{T}$ exactly at the final time.
- (b) **True.** Let $x_0 \in \overleftarrow{\mathcal{R}}_{[t_0, t]}(\mathcal{T})$: there is $\tau \in [t_0, t]$ with $x_{x_0, t_0}(\tau) \in \mathcal{T}$. Because $t < t'$ we have $[t_0, t] \subseteq [t_0, t']$, so the same τ satisfies $\tau \in [t_0, t']$ and $x_{x_0, t_0}(\tau) \in \mathcal{T}$; therefore $x_0 \in \overleftarrow{\mathcal{R}}_{[t_0, t']}(\mathcal{T})$. Backward-reachable tubes are monotone (non-decreasing) in the horizon.
- (c) **False.** The backward-reachable *set* only constrains the state at time t ; it does not care whether the trajectory passed through $\mathcal{F}$ on the way. Counterexample ($n = 1$):

$$\dot{x} = 1, \quad t_0 = 0, \quad t = 2, \quad \mathcal{T} = \{2\}, \quad \mathcal{F} = \{1\}$$

(both nonempty and disjoint). From $x_0 = 0$ the trajectory is $x_{x_0, t_0}(\tau) = \tau$, so $x_{x_0, t_0}(2) = 2 \in \mathcal{T}$ and $x_0 \in \overleftarrow{\mathcal{R}}_2(\mathcal{T})$. However, the only time in $[0, 2]$ at which the trajectory is in $\mathcal{T}$ is $\tau = 2$, and for $s = 1 \in [t_0, \tau]$ we have $x_{x_0, t_0}(1) = 1 \in \mathcal{F}$. The avoid condition fails for the only admissible τ , so $x_0 \notin \overleftarrow{\mathcal{RA}}_{[t_0, 2]}(\mathcal{T}, \mathcal{F})$. Hence $\overleftarrow{\mathcal{R}}_t(\mathcal{T}) \not\subseteq \overleftarrow{\mathcal{RA}}_{[t_0, t]}(\mathcal{T}, \mathcal{F})$ in general.

- (d) **True.** Let $x_0 \in \overleftarrow{\mathcal{RA}}_{[t_0, t]}(\mathcal{T}, \mathcal{F})$. By definition there is $\tau \in [t_0, t]$ such that $x_{x_0, t_0}(\tau) \in \mathcal{T}$ and $x_{x_0, t_0}(s) \notin \mathcal{F}$ for all $s \in [t_0, \tau]$. The first conjunct, with the same witness τ , is precisely the defining condition of $\overleftarrow{\mathcal{R}}_{[t_0, t]}(\mathcal{T})$; hence $x_0 \in \overleftarrow{\mathcal{R}}_{[t_0, t]}(\mathcal{T})$. Dropping a conjunct from a membership condition can only enlarge the set, so the reach-avoid tube is always contained in the reach tube.

Part 2: controlled system $\dot{x} = f(x, u)$

- (a) **True.** Let $x_0 \in \overleftarrow{\mathcal{R}}_{[t_0, t]}^-(\mathcal{T})$: for every $u(\cdot) \in \mathbb{U}_{t_0}^t$ there is $\tau_u \in [t_0, t]$ with $x_{x_0, t_0}^u(\tau_u) \in \mathcal{T}$. Since $\mathbb{U}_{t_0}^t \neq \emptyset$ we may fix any particular $\bar{u}(\cdot) \in \mathbb{U}_{t_0}^t$; then $\tau_{\bar{u}} \in [t_0, t]$ and $x_{x_0, t_0}^{\bar{u}}(\tau_{\bar{u}}) \in \mathcal{T}$, so the pair $(\bar{u}, \tau_{\bar{u}})$ witnesses $x_0 \in \overleftarrow{\mathcal{R}}_{[t_0, t]}^+(\mathcal{T})$. In short, “for all u ” implies “there exists u ” because the set of admissible controls is nonempty: if reaching $\mathcal{T}$ cannot be avoided, then it can certainly be enforced.
- (b) **False.** The enforceable reach-avoid tube is

$$\overleftarrow{\mathcal{RA}}_{[t_0, t]}^+(\mathcal{T}, \mathcal{F}) := \{x_0 \in \mathbb{R}^n : \exists u(\cdot) \in \mathbb{U}_{t_0}^t, \exists \tau \in [t_0, t], x_{x_0, t_0}^u(\tau) \in \mathcal{T} \wedge \forall s \in [t_0, \tau], x_{x_0, t_0}^u(s) \notin \mathcal{F}\}.$$

The intersection $\overleftarrow{\mathcal{R}}_{[t_0, t]}^+(\mathcal{T}) \cap \Omega_{[t_0, t]}(\mathcal{F})$ allows *one* control to reach $\mathcal{T}$ and a *different* control to avoid $\mathcal{F}$, whereas $\overleftarrow{\mathcal{RA}}_{[t_0, t]}^+(\mathcal{T}, \mathcal{F})$ requires a *single* control to do both. Counterexample ($n = 1$):

$$\dot{x} = u, \quad u \in [-1, 1], \quad t_0 = 0, \quad t = 2, \quad \mathcal{T} = \{2\}, \quad \mathcal{F} = \{1\}, \quad x_0 = 0.$$

- $x_0 \in \overleftarrow{\mathcal{R}}_{[t_0, 2]}^+(\mathcal{T})$: with $u \equiv 1$, $x_{x_0, t_0}^u(\tau) = \tau$, so $x_{x_0, t_0}^u(2) = 2 \in \mathcal{T}$.
- $x_0 \in \Omega_{[t_0, 2]}(\mathcal{F})$: with $u \equiv 0$, $x_{x_0, t_0}^u(\tau) \equiv 0 \notin \mathcal{F}$ for all $\tau \in [0, 2]$.
- $x_0 \notin \overleftarrow{\mathcal{RA}}_{[t_0, 2]}^+(\mathcal{T}, \mathcal{F})$: if some control satisfies $x_{x_0, t_0}^u(\tau) = 2$ for some τ , the (continuous) trajectory goes from 0 to 2, so by the intermediate value theorem $x_{x_0, t_0}^u(s) = 1 \in \mathcal{F}$ for some $s \in (0, \tau)$. The avoid condition therefore fails for every candidate pair (u, τ) .

Hence $\overleftarrow{\mathcal{R}}_{[t_0, t]}^+(\mathcal{T}) \cap \Omega_{[t_0, t]}(\mathcal{F}) \not\subseteq \overleftarrow{\mathcal{RA}}_{[t_0, t]}^+(\mathcal{T}, \mathcal{F})$, and the equality is false. (The reverse inclusion fails in general as well, because $\overleftarrow{\mathcal{RA}}_{[t_0, t]}^+(\mathcal{T}, \mathcal{F})$ only asks to avoid $\mathcal{F}$ *until* $\mathcal{T}$ is reached, while $\Omega_{[t_0, t]}(\mathcal{F})$ asks to avoid it over the whole horizon. With $\dot{x} = 1 + u$, $u \in [-\frac{1}{2}, \frac{1}{2}]$, $t_0 = 0$, $t = 10$, $\mathcal{T} = \{1\}$, $\mathcal{F} = \{2\}$, the state $x_0 = 0$ lies in $\overleftarrow{\mathcal{RA}}_{[t_0, 10]}^+(\mathcal{T}, \mathcal{F})$, since $u \equiv 0$ reaches 1 at $\tau = 1$ without touching 2, but not in $\Omega_{[t_0, 10]}(\mathcal{F})$, since $\dot{x} \geq \frac{1}{2}$ forces $x_{x_0, t_0}^u(4) \geq 2$ and thus every trajectory crosses 2 before time $4 < 10$.)

- (c) **False.** Only the inclusion “ $\subseteq$ ” holds in general.

Proof of $\Omega_{[t_0, t]}(\mathcal{F}_1 \cup \mathcal{F}_2) \subseteq \Omega_{[t_0, t]}(\mathcal{F}_1) \cap \Omega_{[t_0, t]}(\mathcal{F}_2)$. If $x_0 \in \Omega_{[t_0, t]}(\mathcal{F}_1 \cup \mathcal{F}_2)$ with witness $u(\cdot)$, then $x_{x_0, t_0}^u(\tau) \notin \mathcal{F}_1 \cup \mathcal{F}_2$ for all $\tau \in [t_0, t]$, i.e. $x_{x_0, t_0}^u(\tau) \notin \mathcal{F}_1$ and $x_{x_0, t_0}^u(\tau) \notin \mathcal{F}_2$. The same $u(\cdot)$ therefore witnesses both $x_0 \in \Omega_{[t_0, t]}(\mathcal{F}_1)$ and $x_0 \in \Omega_{[t_0, t]}(\mathcal{F}_2)$.

Counterexample to “ $\supseteq$ ”. The controls that avoid $\mathcal{F}_1$ and $\mathcal{F}_2$ may be different, and no single control may avoid both. Take $n = 2$, $x = (x_1, x_2)$,

$$\dot{x}_1 = 1, \quad \dot{x}_2 = u, \quad u \in [-1, 1], \quad t_0 = 0, \quad t = 1, \quad x_0 = (0, 0),$$

$$\mathcal{F}_1 = \{(1, x_2) : x_2 \geq 0\}, \quad \mathcal{F}_2 = \{(1, x_2) : x_2 \leq 0\}, \quad \mathcal{F}_1 \cup \mathcal{F}_2 = \{(1, x_2) : x_2 \in \mathbb{R}\}.$$

- $x_0 \in \Omega_{[t_0, 1]}(\mathcal{F}_1)$: with $u \equiv -1$ the trajectory is $(\tau, -\tau)$. It has $x_1 = 1$ only at $\tau = 1$, where $x_2 = -1 < 0$, so it never enters $\mathcal{F}_1$.
- $x_0 \in \Omega_{[t_0, 1]}(\mathcal{F}_2)$: symmetrically, $u \equiv +1$ gives (τ, τ) , which never enters $\mathcal{F}_2$.
- $x_0 \notin \Omega_{[t_0, 1]}(\mathcal{F}_1 \cup \mathcal{F}_2)$: $x_1(\tau) = \tau$ regardless of the control, so at $\tau = 1 \in [0, 1]$ every trajectory satisfies $x_{x_0, t_0}^u(1) = (1, x_2(1)) \in \mathcal{F}_1 \cup \mathcal{F}_2$.

Thus $\Omega_{[t_0,t]}(\mathcal{F}_1) \cap \Omega_{[t_0,t]}(\mathcal{F}_2) \supsetneq \Omega_{[t_0,t]}(\mathcal{F}_1 \cup \mathcal{F}_2)$ in general: being able to dodge each hazard separately does not imply being able to dodge both at once. Taking complements (note $\mathbb{R}^n \setminus \Omega_{[t_0,t]}(\mathcal{F}) = \overleftarrow{\mathcal{R}}_{[t_0,t]}^-(\mathcal{F})$), the inevitable BRT of a union of failure sets can be *strictly larger* than the union of the individual inevitable BRTs; Problem 3 exhibits exactly this effect.

Problem 2: 1-D Reachable Sets

Setup. The control signals $u(\cdot)$ are measurable with values in $[-1, 1]$, and the *all-time* tubes are the tubes over the horizon $[t_0, \infty)$:

$$\begin{aligned}\overleftarrow{\mathcal{R}}_{[t_0,\infty)}^+(\mathcal{M}) &= \{x_0 : \exists u(\cdot), \exists \tau \geq t_0, x_{x_0,t_0}^u(\tau) \in \mathcal{M}\}, \\ \overleftarrow{\mathcal{R}}_{[t_0,\infty)}^-(\mathcal{M}) &= \{x_0 : \forall u(\cdot), \exists \tau_u \geq t_0, x_{x_0,t_0}^u(\tau_u) \in \mathcal{M}\}.\end{aligned}$$

Both tubes contain $\mathcal{M}$ itself (take $\tau = t_0$), and the inevitable tube is contained in the enforceable one (Problem 1.2(a)). Without loss of generality $t_0 = 0$. In every “inevitable” claim below the hitting time is bounded uniformly over the controls, so the answers are unchanged if the all-time tubes are instead defined as $\bigcup_{t \geq t_0}$ of the finite-horizon tubes. The sets of interest are $\mathcal{M}_1 = [\frac{1}{2}, 2]$ and $\mathcal{M}_2 = [-3, -\frac{1}{2}] \cup [2, 3]$.

Part 1: stable dynamics $\dot{x} = -x + u$

Since $\dot{x} \in [-x-1, -x+1]$: for $x > 1$ every control gives $\dot{x} < 0$, for $x < -1$ every control gives $\dot{x} > 0$, and for $x \in [-1, 1]$ the controller can hold the state ($u \equiv x$), push it up ($u \equiv 1$) or push it down ($u \equiv -1$). The extreme closed-loop trajectories are

$$\begin{aligned}u \equiv 1 : \quad x(t) &= 1 + (x_0 - 1)e^{-t} \quad (\text{monotone to } 1), \\ u \equiv -1 : \quad x(t) &= -1 + (x_0 + 1)e^{-t} \quad (\text{monotone to } -1),\end{aligned}$$

and by the comparison principle every trajectory is sandwiched between them:

$$-1 + (x_0 + 1)e^{-t} \leq x(t) \leq 1 + (x_0 - 1)e^{-t}, \quad \text{hence} \quad \min(x_0, -1) \leq x(t) \leq \max(x_0, 1). \quad (1)$$

The drift pulls every state toward $[-1, 1]$ and no control can move the state *away* from the origin past ± 1 .

(a) $\mathcal{M}_1 = [\frac{1}{2}, 2]$.

Enforceable tube: $\overleftarrow{\mathcal{R}}_{[t_0,\infty)}^+(\mathcal{M}_1) = \mathbb{R}$. For $x_0 \in \mathcal{M}_1$ take $\tau = 0$. For $x_0 > 2$, every control gives $\dot{x} \leq -x + 1 \leq -1$ as long as $x \geq 2$, so the state decreases and hits $2 \in \mathcal{M}_1$ no later than $\tau = x_0 - 2$ (for *every* control). For $x_0 < \frac{1}{2}$, the control $u \equiv 1$ gives $x(t) = 1 + (x_0 - 1)e^{-t} \uparrow 1$, which crosses $\frac{1}{2}$ at the finite time $\tau = \ln(2(1 - x_0))$.

Inevitable tube: $\overleftarrow{\mathcal{R}}_{[t_0,\infty)}^-(\mathcal{M}_1) = [\frac{1}{2}, \infty)$. Points of $\mathcal{M}_1$ and points $x_0 > 2$ are inevitable by the argument above. For $x_0 < \frac{1}{2}$ the control $u \equiv -1$ gives $x(t) = -1 + (x_0 + 1)e^{-t}$, which stays between x_0 and -1 for all times, hence $x(t) \leq \max(x_0, -1) < \frac{1}{2}$ and the trajectory never enters $\mathcal{M}_1$.

(b) $\mathcal{M}_2 = [-3, -\frac{1}{2}] \cup [2, 3]$.

Enforceable tube: $\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^+(\mathcal{M}_2) = \mathbb{R}$. For $x_0 \in \mathcal{M}_2$ take $\tau = 0$. For $x_0 > 3$, every control gives $\dot{x} \leq -x + 1 \leq -2$ while $x \geq 3$, so the state hits $3 \in \mathcal{M}_2$ no later than $\tau = (x_0 - 3)/2$; symmetrically, for $x_0 < -3$ every control gives $\dot{x} \geq -x - 1 \geq 2$ while $x \leq -3$, so the state hits $-3 \in \mathcal{M}_2$ no later than $\tau = (-3 - x_0)/2$. For $x_0 \in (-\frac{1}{2}, 2)$ the control $u \equiv -1$ gives $x(t) = -1 + (x_0 + 1)e^{-t} \downarrow -1$, which crosses $-\frac{1}{2}$ at the finite time $\tau = \ln(2(x_0 + 1))$.

Inevitable tube: $\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^-(\mathcal{M}_2) = (-\infty, -\frac{1}{2}] \cup [2, \infty)$. Points of $\mathcal{M}_2$ and points with $x_0 > 3$ or $x_0 < -3$ are inevitable by the argument above. The remaining points $x_0 \in (-\frac{1}{2}, 2)$ can avoid $\mathcal{M}_2$ forever: if $x_0 \in (-\frac{1}{2}, 1]$, the constant control $u \equiv x_0$ (admissible since $|x_0| \leq 1$) holds $x(t) \equiv x_0 \notin \mathcal{M}_2$; if $x_0 \in (1, 2)$, the control $u \equiv 1$ gives $x(t) = 1 + (x_0 - 1)e^{-t} \in (1, x_0] \subset (1, 2)$ for all times.

Part 2: unstable dynamics $\dot{x} = x + u$

Now $\dot{x} \in [x - 1, x + 1]$: for $x > 1$ every control gives $\dot{x} > 0$, for $x < -1$ every control gives $\dot{x} < 0$, and on $(-1, 1)$ the controller can hold the state ($u \equiv -x$) or steer it in either direction. The extreme trajectories are

$$u \equiv 1 : x(t) = -1 + (x_0 + 1)e^t, \quad u \equiv -1 : x(t) = 1 + (x_0 - 1)e^t,$$

and every trajectory satisfies

$$1 + (x_0 - 1)e^t \leq x(t) \leq -1 + (x_0 + 1)e^t. \quad (2)$$

Consequently: if $x_0 > 1$ then $x(t) > x_0$ for all $t > 0$ and $x(t) \rightarrow \infty$ for every control; if $x_0 \leq -1$ then $x(t) \leq -1$ for all times (and $x(t) \rightarrow -\infty$ if $x_0 < -1$); the drift now pushes the state *away* from the origin, and the controller only has authority inside $[-1, 1]$.

(a) $\mathcal{M}_1 = [\frac{1}{2}, 2]$.

Enforceable tube: $\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^+(\mathcal{M}_1) = (-1, 2]$. For $x_0 \in \mathcal{M}_1$ take $\tau = 0$. For $x_0 \in (-1, \frac{1}{2})$ the control $u \equiv 1$ gives $x(t) = -1 + (x_0 + 1)e^t \uparrow \infty$, which crosses $\frac{1}{2}$ at the finite time $\tau = \ln(\frac{3}{2(x_0 + 1)})$. For $x_0 > 2$, (2) gives $x(t) > x_0 > 2$ for every control, and for $x_0 \leq -1$ it gives $x(t) \leq -1 < \frac{1}{2}$ for every control; neither can ever reach $\mathcal{M}_1$.

Inevitable tube: $\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^-(\mathcal{M}_1) = [\frac{1}{2}, 2] = \mathcal{M}_1$. Points outside $(-1, 2]$ are not even enforceable. For $x_0 \in (-1, \frac{1}{2})$ the control $u \equiv -1$ gives $x(t) = 1 + (x_0 - 1)e^t$, which is decreasing because $x_0 < 1$, so $x(t) \leq x_0 < \frac{1}{2}$ forever and $\mathcal{M}_1$ is avoided.

(b) $\mathcal{M}_2 = [-3, -\frac{1}{2}] \cup [2, 3]$.

Enforceable tube: $\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^+(\mathcal{M}_2) = [-3, 3]$. For $x_0 \in \mathcal{M}_2$ take $\tau = 0$. For $x_0 \in (-\frac{1}{2}, 2)$ the control $u \equiv 1$ gives $x(t) = -1 + (x_0 + 1)e^t \uparrow \infty$, which crosses $2 \in \mathcal{M}_2$ in finite time. For $x_0 > 3$ we have $x(t) > x_0 > 3$, and for $x_0 < -3$ we have $x(t) < x_0 < -3$, for every control and all $t > 0$; $\mathcal{M}_2$ is never reached.

Inevitable tube: $\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^-(\mathcal{M}_2) = [-3, -\frac{1}{2}] \cup (1, 3]$. For $x_0 \in (1, 2)$, (2) gives $x(t) \geq 1 + (x_0 - 1)e^t$ for every control; the right-hand side equals 2 at $t = \ln(1/(x_0 - 1))$, so the continuous trajectory, which starts below 2, crosses $2 \in \mathcal{M}_2$ no later than that time. Thus $(1, 2) \subset \overleftarrow{\mathcal{R}}_{[t_0, \infty)}^-(\mathcal{M}_2)$ even

though $(1,2) \cap \mathcal{M}_2 = \emptyset$: the unstable drift drags these states into $[2,3]$. For $x_0 \in (-\frac{1}{2}, 1]$ the constant control $u \equiv -x_0$ (admissible since $|x_0| \leq 1$) holds $x(t) \equiv x_0 \notin \mathcal{M}_2$. Points with $|x_0| > 3$ are not enforceable.

	stable $\dot{x} = -x + u$	unstable $\dot{x} = x + u$
$\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^+(\mathcal{M}_1)$	$\mathbb{R}$	$(-1, 2]$
$\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^-(\mathcal{M}_1)$	$[\frac{1}{2}, \infty)$	$[\frac{1}{2}, 2]$
$\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^+(\mathcal{M}_2)$	$\mathbb{R}$	$[-3, 3]$
$\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^-(\mathcal{M}_2)$	$(-\infty, -\frac{1}{2}] \cup [2, \infty)$	$[-3, -\frac{1}{2}] \cup (1, 3]$

Table 1: Summary of the all-time backward-reachable tubes in Problem 2.

Part 3: a set that is strictly easier to reach with the unstable dynamics

Take $\mathcal{M}_3 := [2, \infty)$.

- *Stable dynamics:* $\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^+(\mathcal{M}_3) = [2, \infty) = \mathcal{M}_3$. Points of $\mathcal{M}_3$ are trivially included. For $x_0 < 2$, (1) gives $x(t) \leq \max(x_0, 1) < 2$ for every control and every time, so $\mathcal{M}_3$ is never reached: bounded controls cannot overcome the drift toward the origin once $x > 1$.
- *Unstable dynamics:* $\overleftarrow{\mathcal{R}}_{[t_0, \infty)}^-(\mathcal{M}_3) = (-1, \infty)$. For $x_0 > -1$ the control $u \equiv 1$ gives $x(t) = -1 + (x_0 + 1)e^t \uparrow \infty$, which crosses 2 in finite time. For $x_0 \leq -1$, (2) gives $x(t) \leq -1$ for every control, so $\mathcal{M}_3$ is unreachable.

Hence $(-1, \infty) \supsetneq [2, \infty)$: the enforceable tube under the unstable dynamics strictly contains the one under the stable dynamics. Intuitively, the controller can nudge any state in $(-1, 1)$ slightly past 1 and then let the instability carry it arbitrarily far, whereas the stable drift pulls everything back toward $[-1, 1]$, so a far-away target can only be reached from states that are already beyond it. Any $\mathcal{M}_3 = [a, \infty)$ with $a \geq 1$ works (as does the symmetric $(-\infty, -a] \cup [a, \infty)$, whose enforceable tubes are itself and $\mathbb{R}$, respectively). Bounded targets cannot work: for $\mathcal{M}_3 = [a, b]$ with $a > 1$ the two enforceable tubes are $[a, \infty)$ and $(-1, b]$, and neither contains the other, since the stable drift lets far-away states fall back onto the target while the unstable drift never lets states beyond the target come back.

Problem 3: Backward Reachable Set – Programming

Implementation

The robot advances one row per step and can shift by at most one column, $(x_{t+1}, y_{t+1}) = (x_t + u_t, y_t + 1)$ with $u_t \in \{-1, 0, 1\}$; landing on an obstacle cell is a failure. `get_successors` returns the cells reachable in one step (moves that would leave the grid are treated as inadmissible; for the obstacle layout in this problem the tube never touches the side walls, so this choice does not affect the results). `calculate_inevitable_brt` computes the inevitable BRT of the failure set $\mathcal{F}$ by the fixed-point iteration

$$\overleftarrow{\mathcal{R}}_0^- = \mathcal{F}, \quad \overleftarrow{\mathcal{R}}_{k+1}^- = \overleftarrow{\mathcal{R}}_k^- \cup \{x : f(x, u) \in \overleftarrow{\mathcal{R}}_k^- \text{ for all } u \in \{-1, 0, 1\}\},$$

i.e. a cell is added when every admissible control leads into the current set (no escape route), and the iteration stops when a sweep adds nothing. Because y increases by one every step, only the rows below the highest obstacle row can ever collide, and the loop converges after at most that many sweeps. The complement of the resulting set is the (all-time) safe set $\Omega(\mathcal{F})$; this is the discrete-time, finite-state counterpart of the HJ reachability formulation in [1].

```

1 def get_successors(cell, grid_size):
2     """Calculates the three successor states based on the forward dynamics."""
3     x, y = cell
4     successors = []
5     # Dynamics:  $y_{t+1} = y_t + 1$  and  $x_{t+1} = x_t + u_t$  with  $u_t$  in  $\{-1, 0, 1\}$ .
6     # A move that would leave the grid is not admissible, so a cell on the
7     # left/right border has only two successors (this never matters for the
8     # obstacle layout in this problem, since the BRT stays away from the walls).
9     for u in (-1, 0, 1):
10         x_next, y_next = x + u, y + 1
11         if 0 <= x_next < grid_size and 0 <= y_next < grid_size:
12             successors.append((x_next, y_next))
13     # successors is a list of tuples. each tuple is a (x,y) coordinate pair.
14     return successors

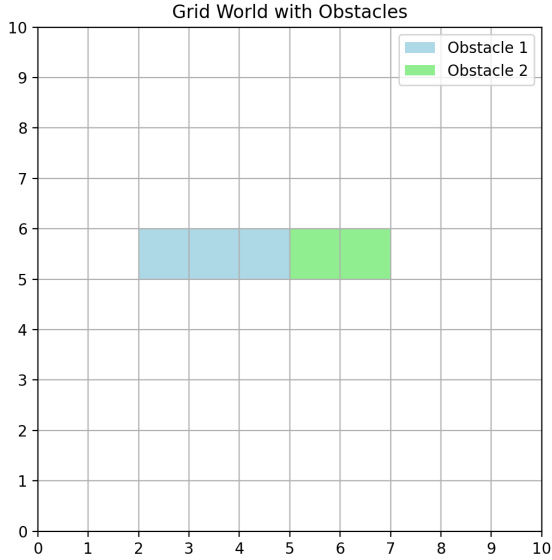
1 def calculate_inevitable_brt(obstacles, grid_size):
2     """Calculates the set of states from which eventual collision is unavoidable."""
3     inevitable_brt = set(obstacles)
4     while True:
5         newly_added_states = set()
6         max_obs_y = max(y for x, y in obstacles) if obstacles else -1
7         # check all cells below this y-level
8         # ensure that for any given cell that you check, all successors must
9         # hit the obstacles (otherwise there is an escape route and a collision
10        # is not inevitable)
11        for y in range(max_obs_y):          # rows y = 0, ..., max_obs_y - 1
12            for x in range(grid_size):
13                cell = (x, y)
14                if cell in inevitable_brt:
15                    continue
16                successors = get_successors(cell, grid_size)
17                # Every admissible control leads into the current tube, i.e.
18                # there is no escape route, so collision is inevitable.
19                if successors and all(s in inevitable_brt for s in successors):
20                    newly_added_states.add(cell)
21            if not newly_added_states:
22                break          # fixed point reached
23        inevitable_brt |= newly_added_states
24
25    # Your brt should be a set of tuples. Each tuple is a (x,y) coordinate pair.
26    return inevitable_brt

```

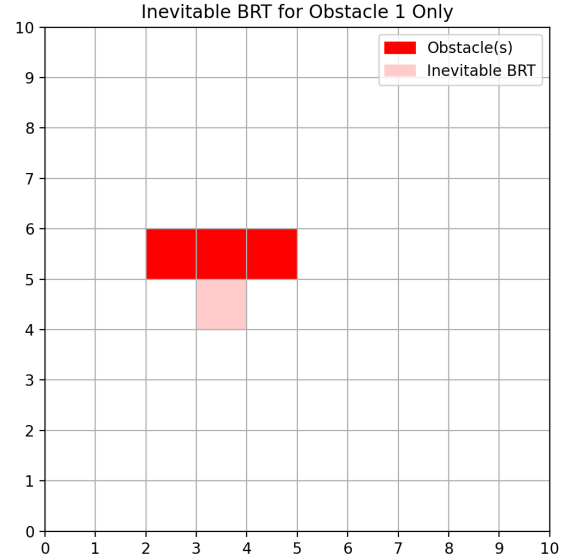
Results

Figure 1 shows the grid world and the three inevitable BRTs produced by the notebook. With obstacle 1 occupying columns $\{2, 3, 4\}$ of row 5 and obstacle 2 occupying columns $\{5, 6\}$:

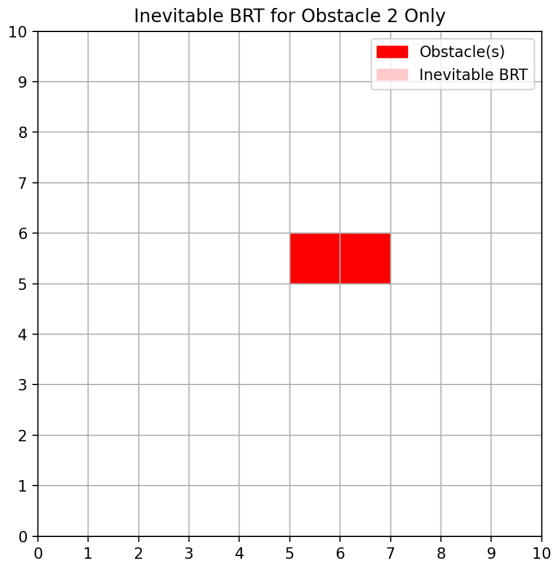
failure set	inevitable BRT minus the obstacle cells
obstacle 1 (width 3)	$\{(3, 4)\}$
obstacle 2 (width 2)	$\emptyset$
both obstacles (width 5)	$\{(3, 4), (4, 4), (5, 4), (4, 3)\}$



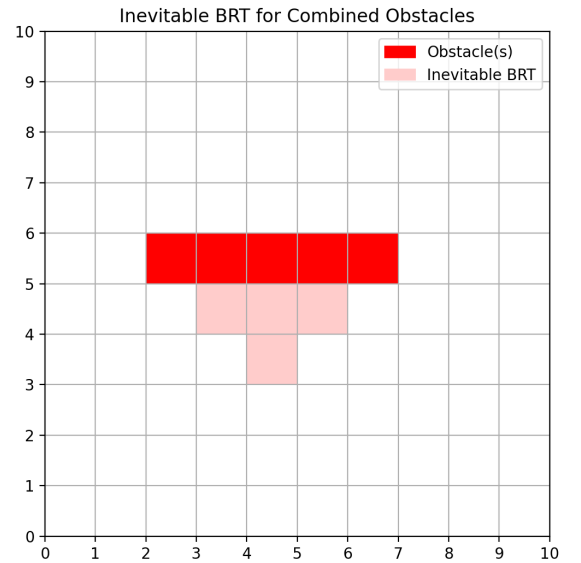
(a) The grid world.



(b) Obstacle 1 only.



(c) Obstacle 2 only.



(d) Both obstacles.

Figure 1: Inevitable backward-reachable tubes (light red) of the obstacle sets (red).

Discussion

Why the tubes look the way they do. The robot cannot stop or slow down, so a cell d rows below the obstacle row will land on that row after exactly d steps, at a column anywhere in $\{x - d, \dots, x + d\}$ (any integer in that range is attainable by an appropriate control sequence). Collision is inevitable if and only if *every* one of those landing columns is occupied, i.e. $\{x - d, \dots, x + d\} \subseteq \text{obstacle columns}$. For a contiguous obstacle of width w this means $2d \leq w - 1$, so the tube is an inverted triangle (a cone with unit slope) of height $\lfloor (w - 1)/2 \rfloor$ below the obstacle's centre: height 1 for the width-3 obstacle, height 0 for the width-2 obstacle, and height 2 for the combined width-5 obstacle. This is exactly what the code produces, and it matches intuition: since the robot's lateral speed equals its forward speed, it can escape any obstacle whose half-width is smaller than its current distance to the obstacle row, so the doomed region hugs the obstacle tightly. Everything above the obstacle row and everything in the rows below it that is outside the cone is safe. Note that the width-2 obstacle alone has an empty tube beyond itself: from (5, 4) or (6, 4) the robot can always step to a free neighbouring column.

Combined obstacles versus the union of the individual tubes. The union of the two individual tubes adds only (3, 4) to the obstacle cells, whereas the tube of the combined failure set adds (3, 4), (4, 4), (5, 4) and (4, 3) (Figure 2). The combined tube *strictly contains* the union. The extra cells are states from which every escape from one obstacle lands in the other: from (4, 4), steering left or going straight hits obstacle 1 while steering right hits obstacle 2, and from (4, 3) all three successors are among these doomed cells. This is the discrete version of Problem 1.2(c): a state can be safe with respect to $\mathcal{F}_1$ and safe with respect to $\mathcal{F}_2$ using two *different* controllers, yet unsafe with respect to $\mathcal{F}_1 \cup \mathcal{F}_2$, i.e. $\Omega(\mathcal{F}_1 \cup \mathcal{F}_2) \subsetneq \Omega(\mathcal{F}_1) \cap \Omega(\mathcal{F}_2)$ and, equivalently, $\overleftarrow{\mathcal{R}}^-(\mathcal{F}_1 \cup \mathcal{F}_2) \supsetneq \overleftarrow{\mathcal{R}}^-(\mathcal{F}_1) \cup \overleftarrow{\mathcal{R}}^-(\mathcal{F}_2)$. The practical lesson is that safety analysis must be carried out on the *full* failure set; per-obstacle tubes cannot simply be unioned, and a “harmless” obstacle (obstacle 2 alone traps nothing) can enlarge the unsafe region substantially once it sits next to another one. The boundary of the tube is also where a least-restrictive safety filter would have to intervene: from (3, 3) the only safe action is $u = -1$, so the filter must override $u \in \{0, +1\}$ there, while it can leave the controller free anywhere outside the tube.

Effect of the forward-to-lateral speed ratio. The shape of the tube is set by the ratio of lateral to forward speed. If the robot advanced v rows per step while still shifting by at most one column (equivalently, it could steer only once every v rows), then from d rows away it could only reach columns within $\pm d/v$ of its current one, and the trapping condition becomes $2d/v \leq w - 1$: the cone becomes steeper and its height grows to about $v(w - 1)/2$, so the inevitable tube *grows*. A faster robot has fewer steps in which to swerve, and in the limit of very fast forward motion every cell in the obstacle's “shadow” directly below it becomes doomed. Conversely, more lateral authority ($u \in \{-k, \dots, k\}$) flattens the cone to height $\lfloor (w - 1)/(2k) \rfloor$; with $k = 2$ the combined tube would shrink to the obstacles plus the single cell (4, 4), and obstacle 1's tube would shrink to the obstacle itself. (In the literal discrete model a speed of $v = 2$ rows per step also introduces a parity effect: a robot starting on an even row would hop over the single obstacle row entirely, since grazing is allowed. The statement above should therefore be read in the fine-grid or continuous-time limit, where the obstacle has finite thickness.)

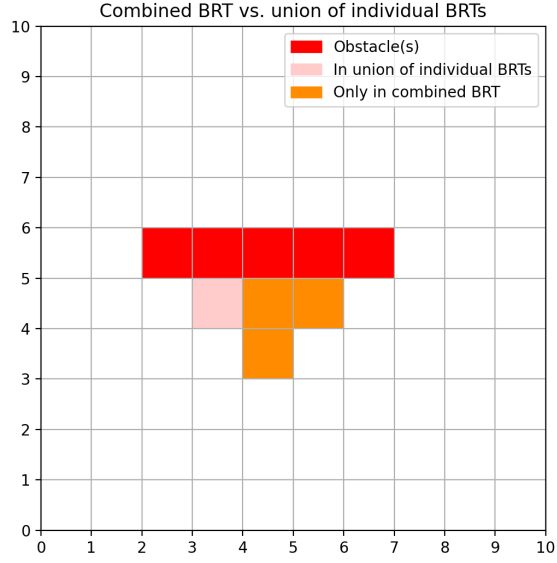


Figure 2: The tube of the combined failure set. Light red cells also belong to the union of the two individual tubes; orange cells belong only to the combined tube.

References

- [1] S. Bansal, M. Chen, S. Herbert, and C. J. Tomlin, "Hamilton-Jacobi Reachability: A Brief Overview and Recent Advances," in *2017 IEEE 56th Annual Conference on Decision and Control (CDC)*, 2017, pp. 2242–2253.